

Memory Maze - Human Vs Ai Pathfinding Game Design Document

Version 1.1



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Revision History

Date	Version	Description	Author
12 / 01 / 2026	1.0	Introduction	BC220407371
11 / 05 / 2026	1.1	Finalizing	BC220407371

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Introduction to Design Document

This Design Document describes the detailed system design of “Memory Maze – Human vs AI Pathfinding Game”, developed as a Final Year Project. It explains how the requirements defined in

the Software Requirements Specification (SRS) are converted into a structured and implementable system design.

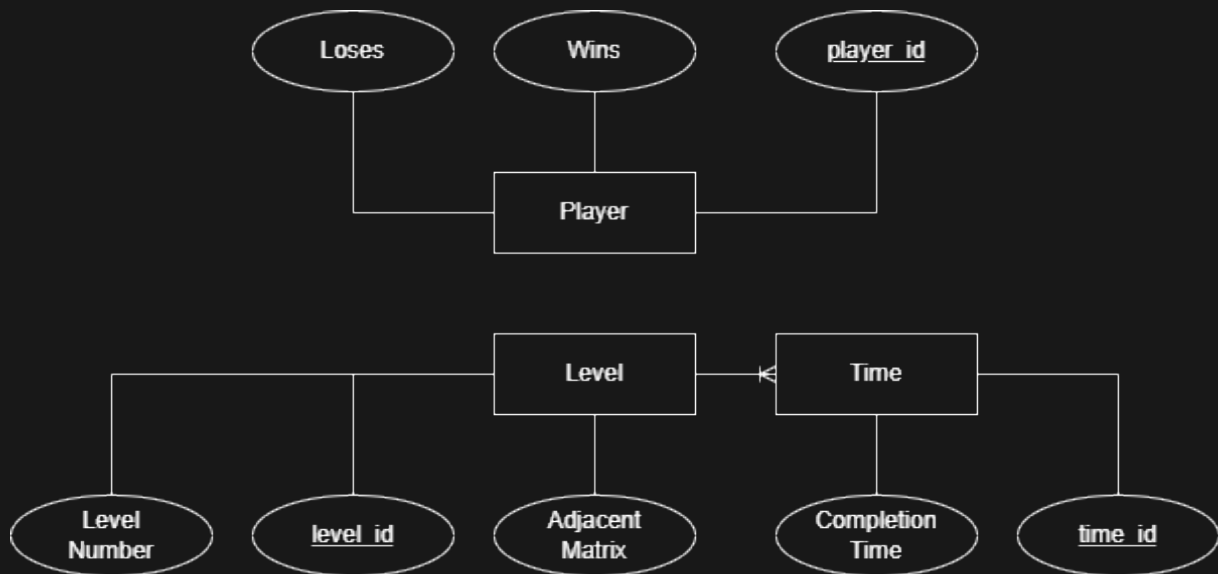
This document includes the

- Entity Relationship Diagram (ERD) to define the data structure
- Sequence diagrams to illustrate interactions among system components
- An architecture design diagram to present the overall system organization
- Class diagrams to represent the object-oriented structure of the system.
- Database design for data storage
- Interface design to describe user interaction
- Test cases to validate the correctness of the proposed design

The purpose of the design phase is to establish a clear blueprint for development by identifying system components, defining their responsibilities, and specifying interactions before implementation begins. This phase helps reduce ambiguity, detect design issues early, and improve system maintainability through a modular and organized structure.

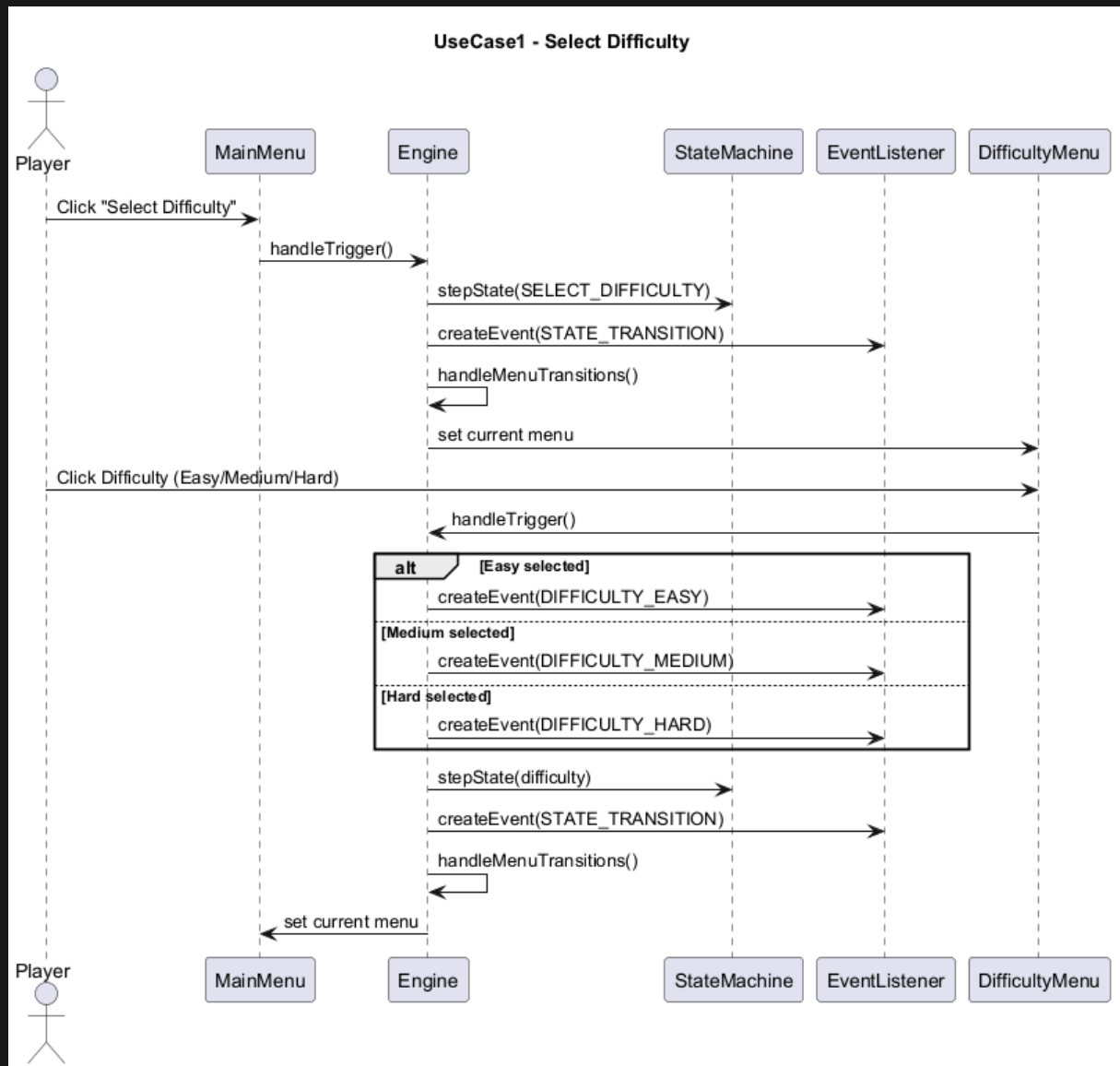
By providing visual and structural representations of the system, the design phase enhances communication among developers and evaluators and ensures efficient and reliable implementation of “Memory Maze – Human vs AI Pathfinding Game”.

Entity Relationship Diagram

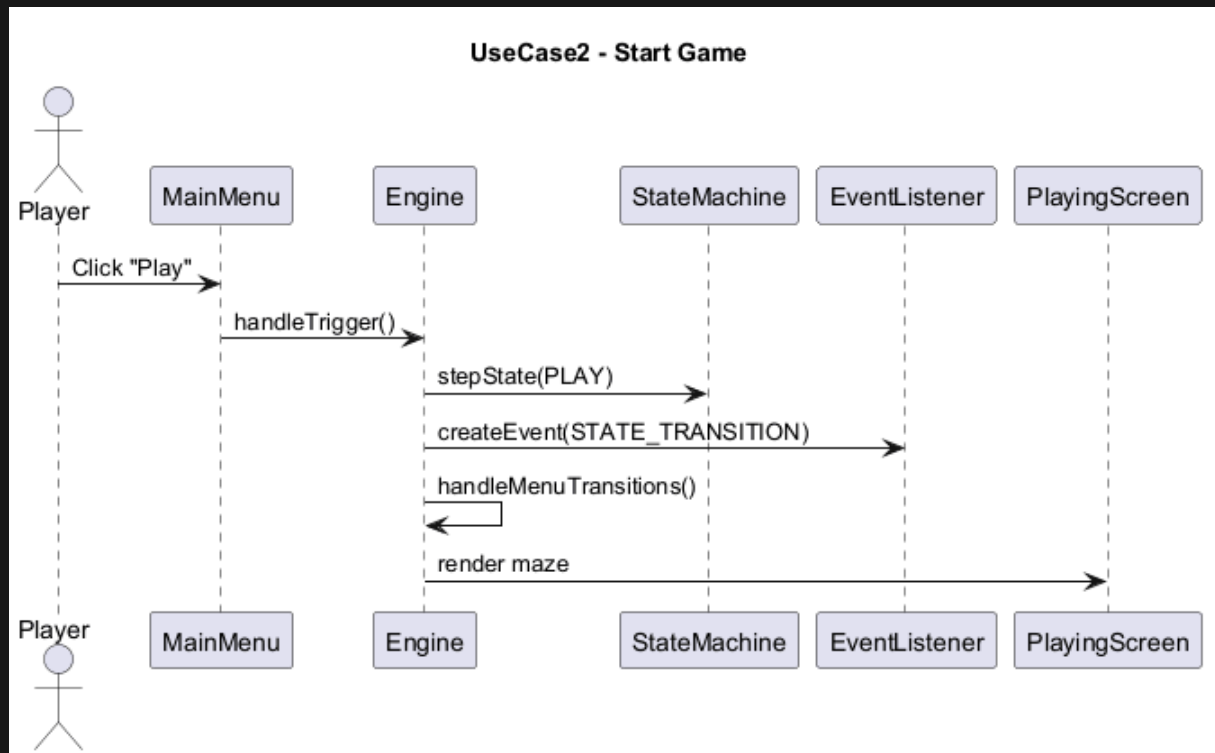


Sequence Diagrams

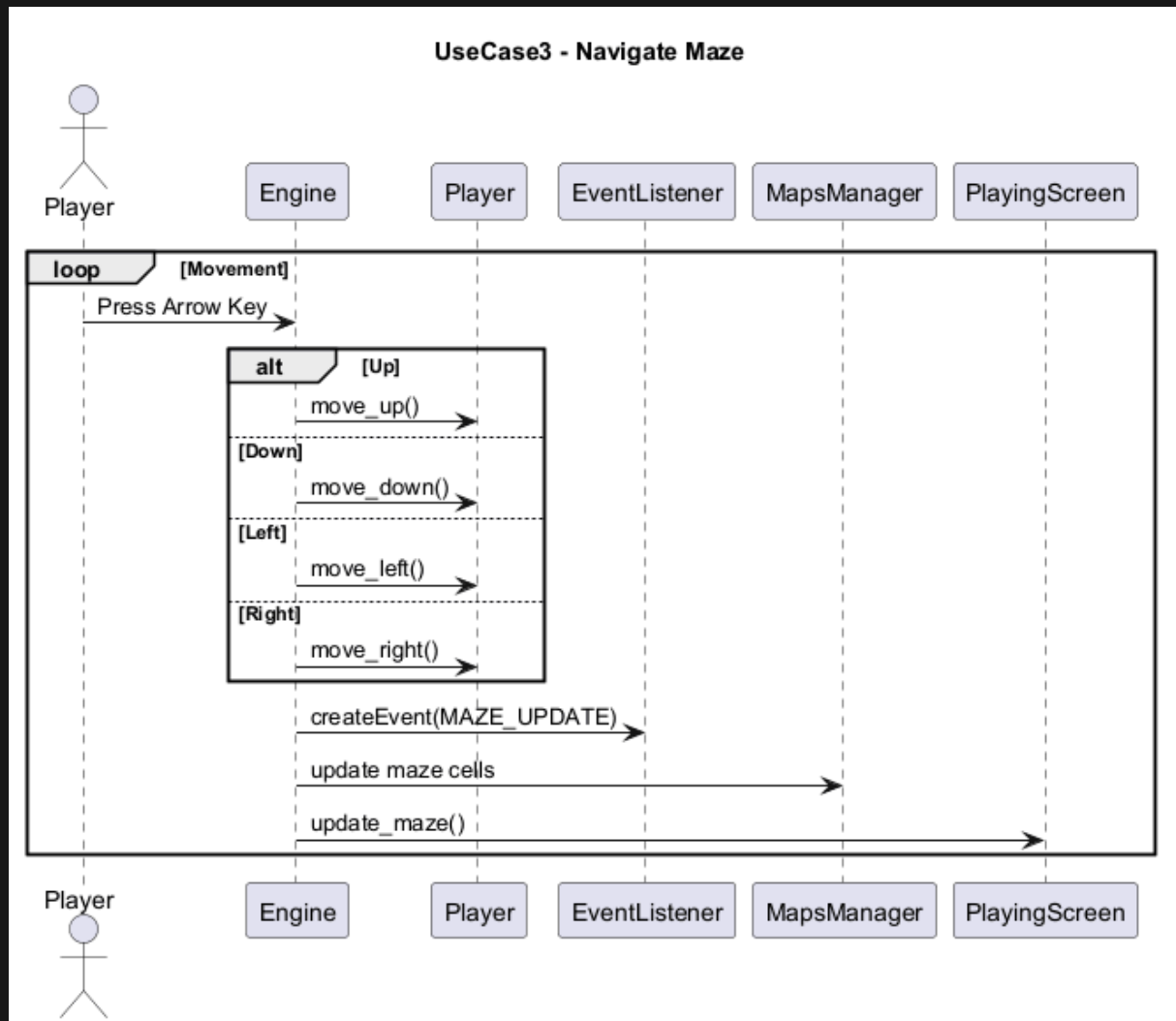
Select Difficulty



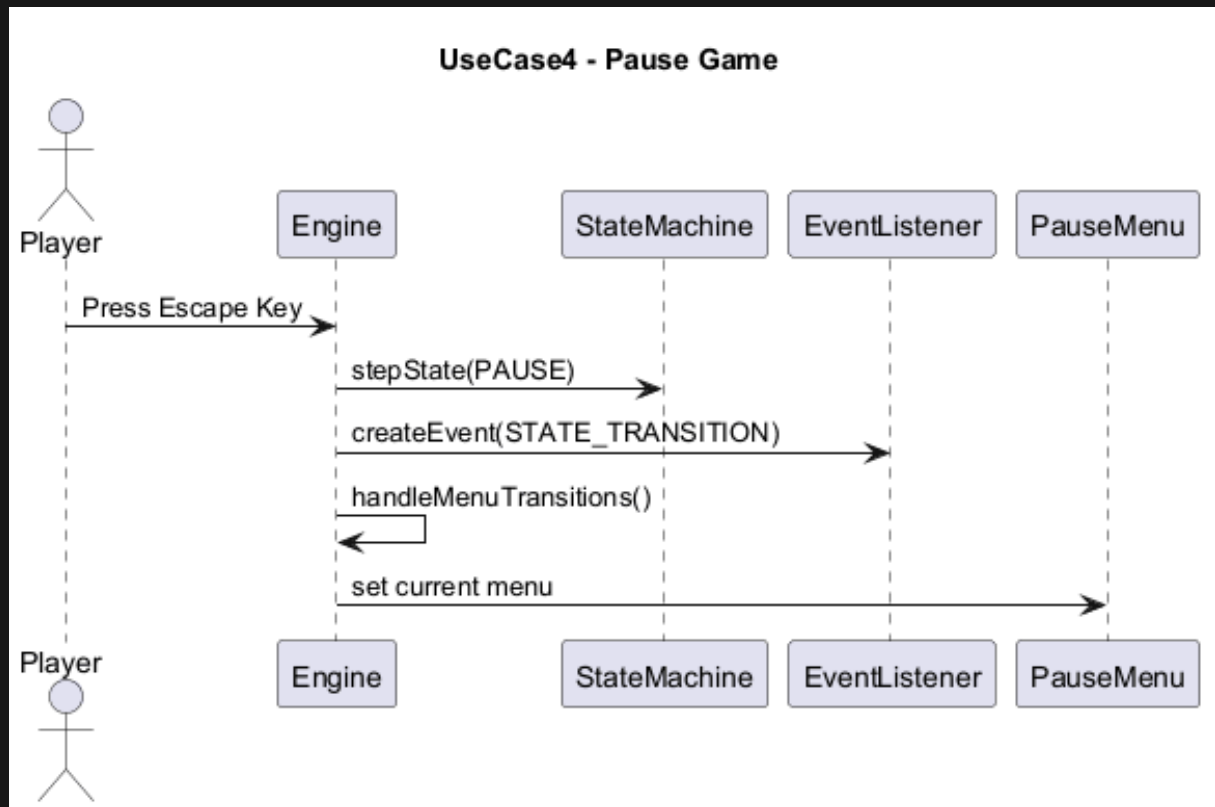
Start Game



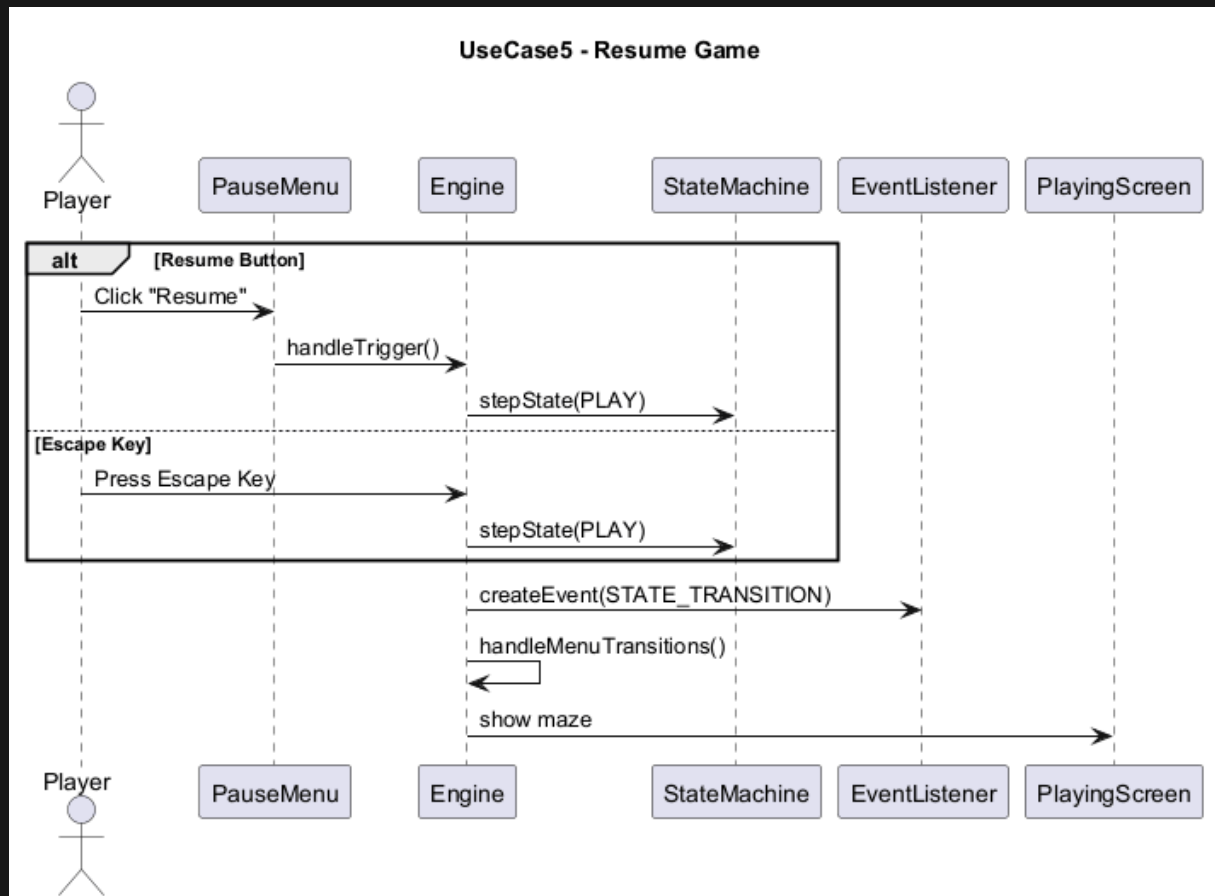
Navigate Maze



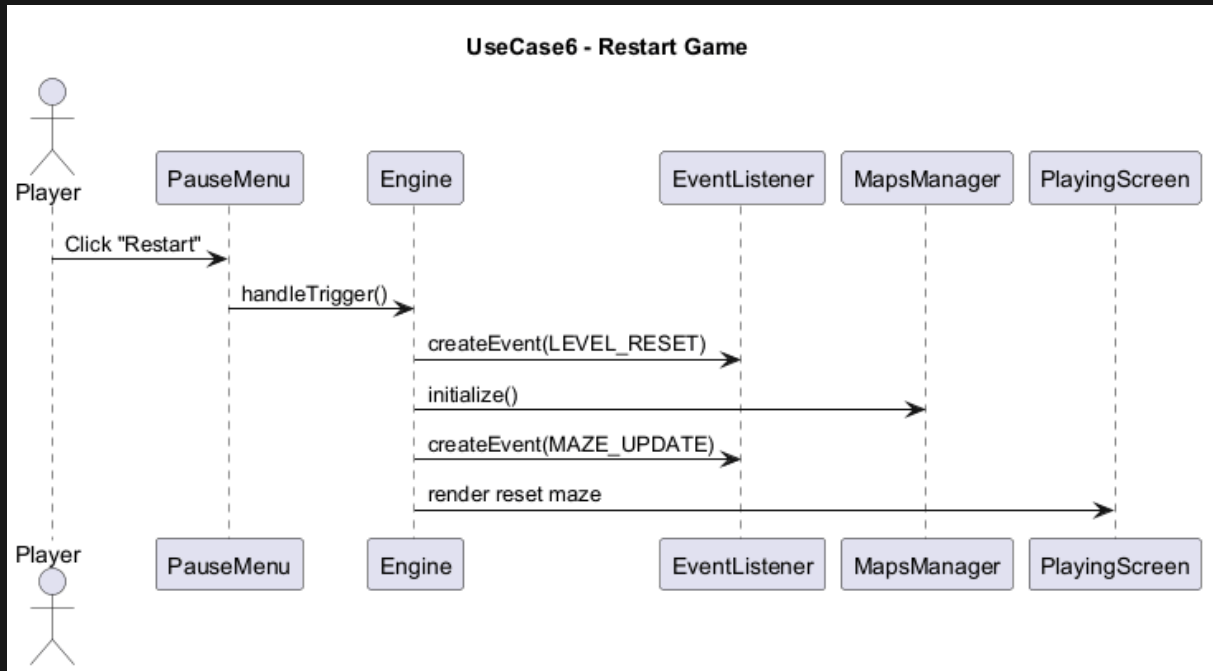
Pause Game



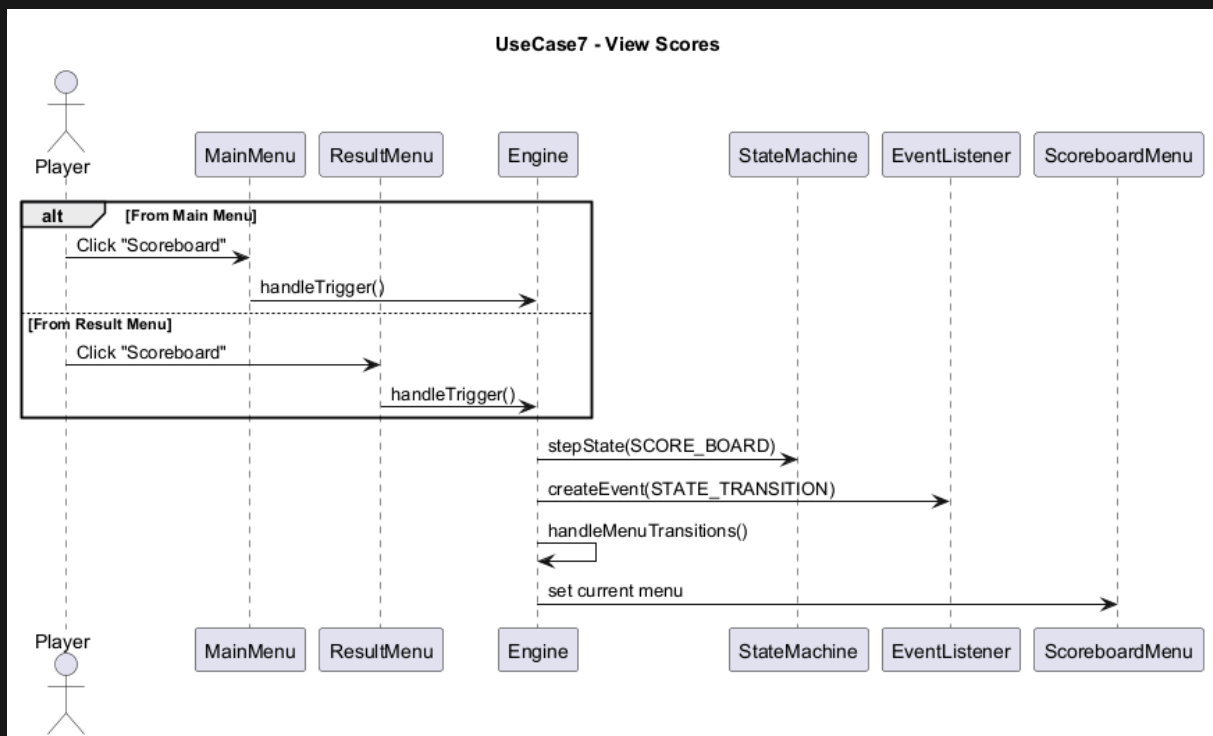
Resume Game



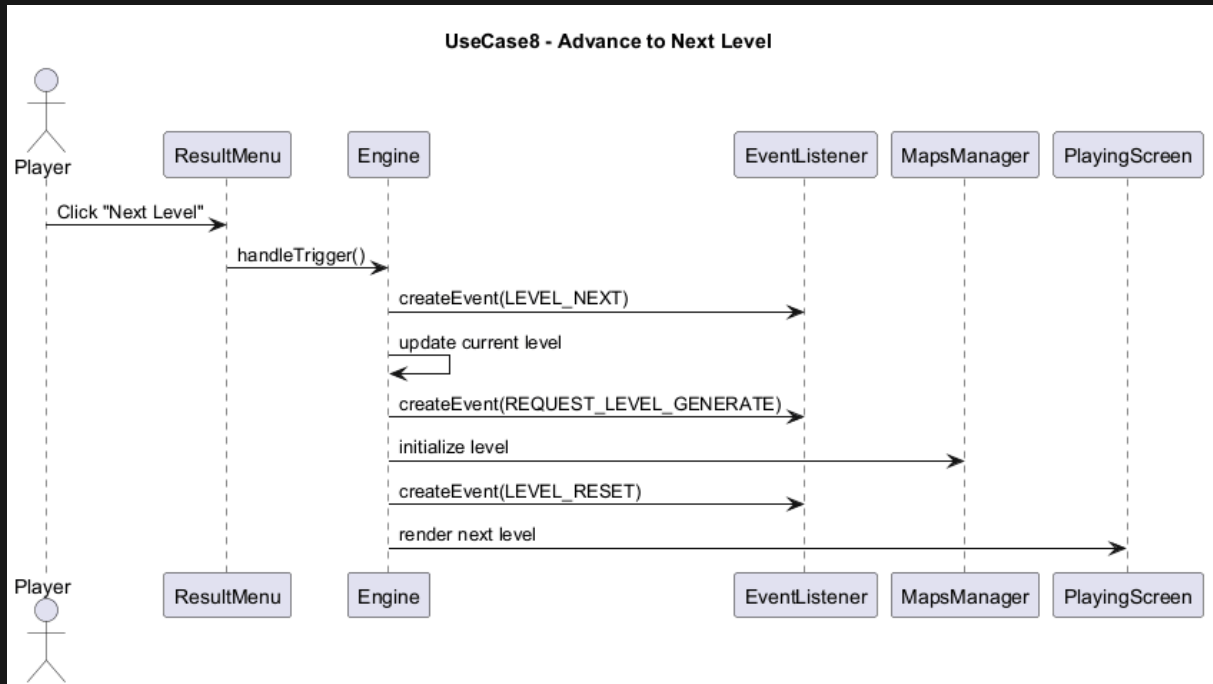
Restart Game



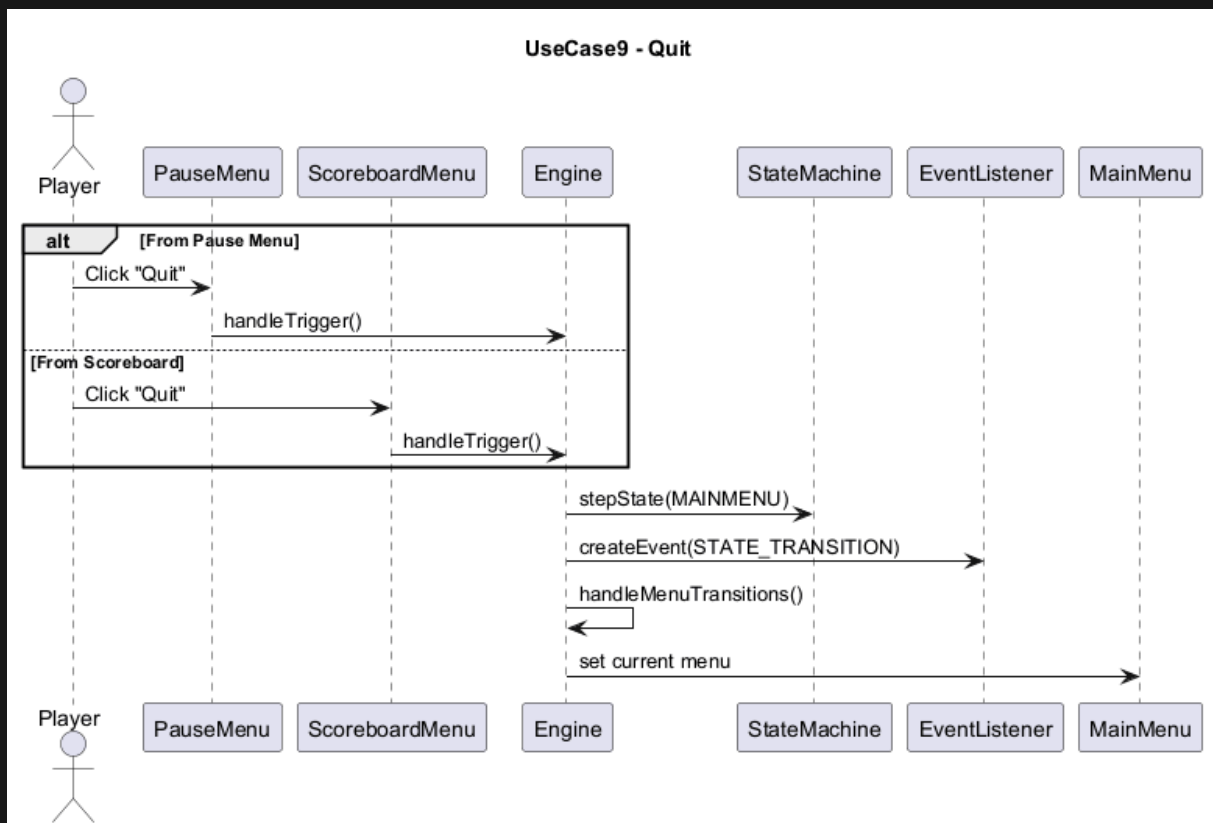
View Scores



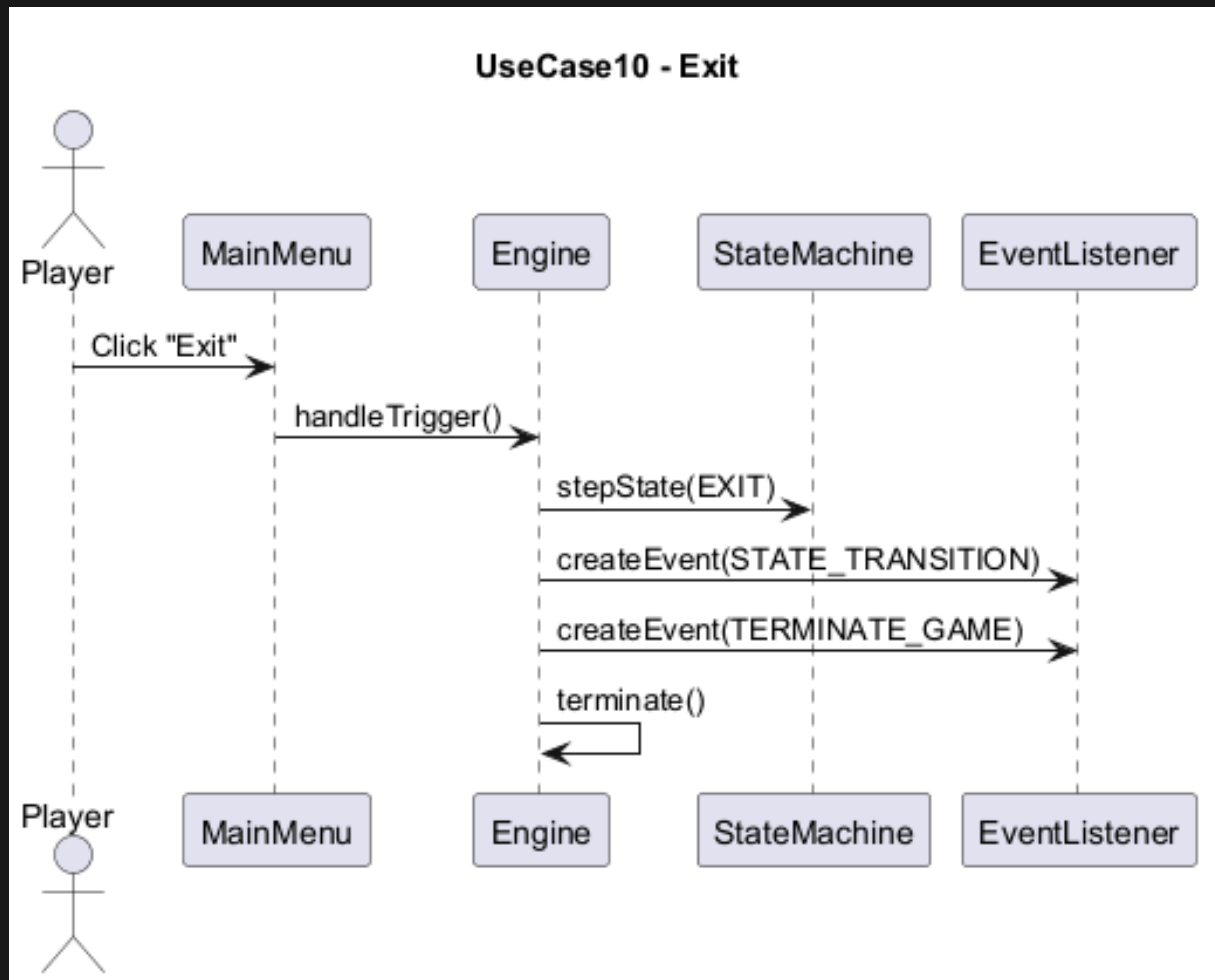
Advance to next Level



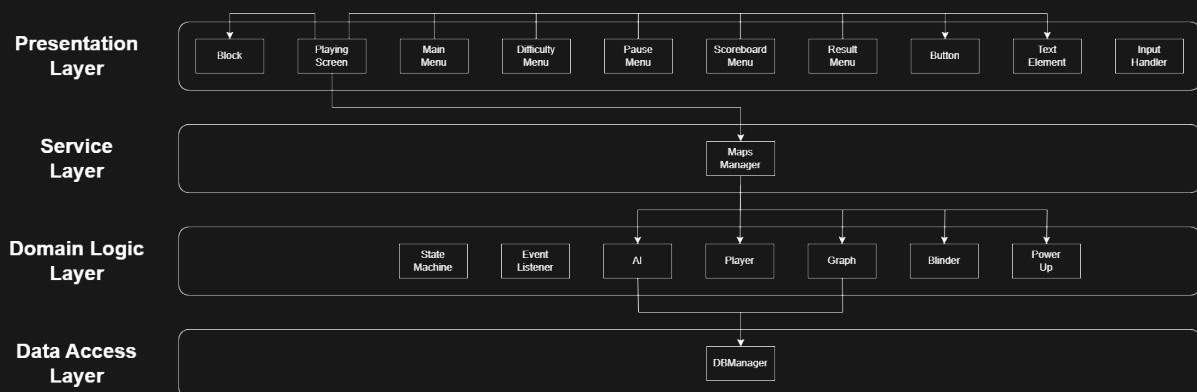
Quit



Exit



Architecture Design Diagram



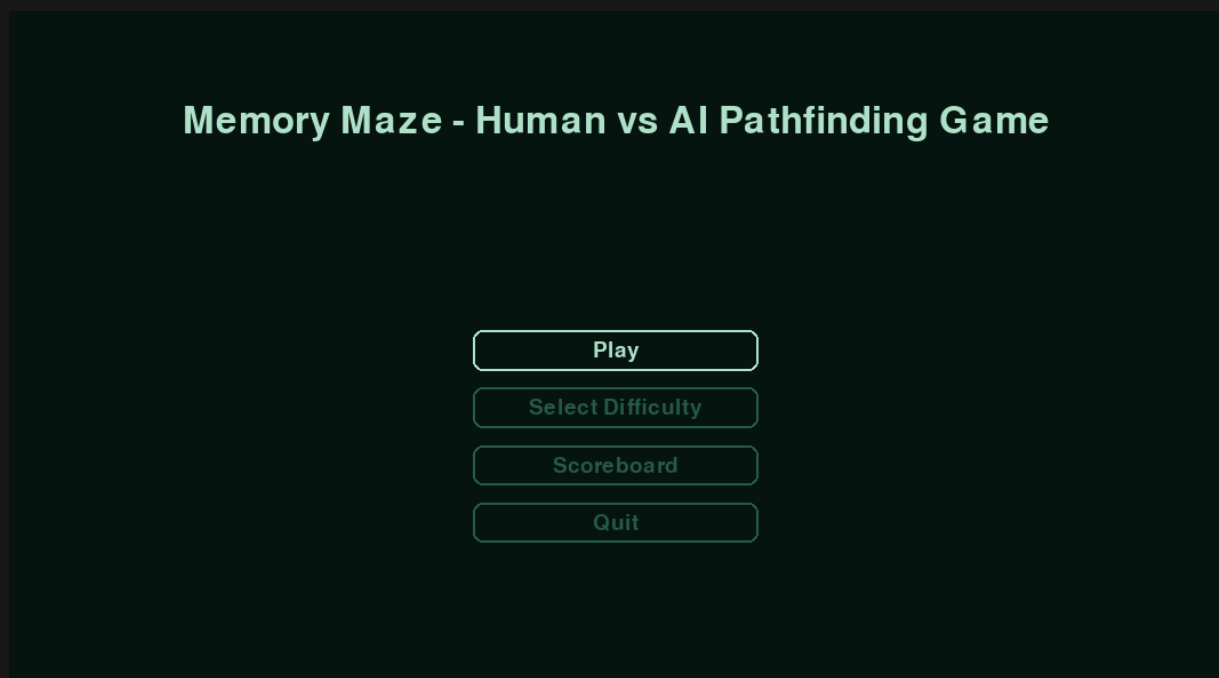
Class Diagram

Database Design



Interface Design

Main Menu



Difficulty Menu

Difficulty Menu

Easy

Medium

Hard

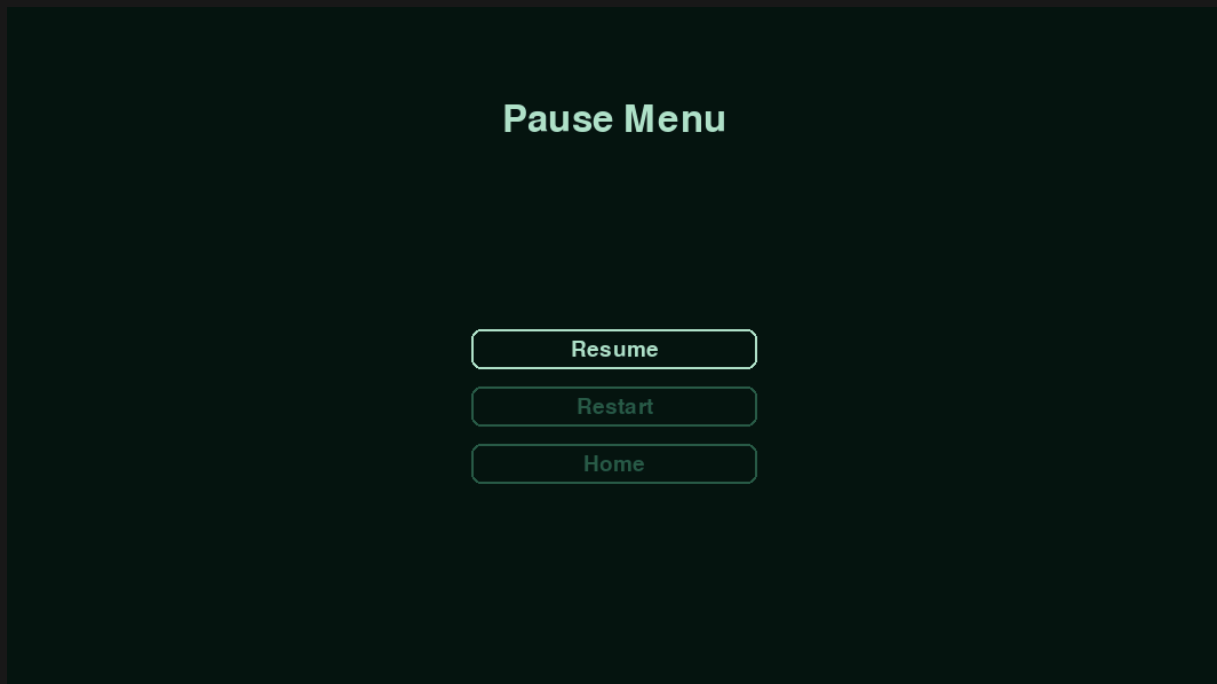
Playing Screen

Timer: 00:06

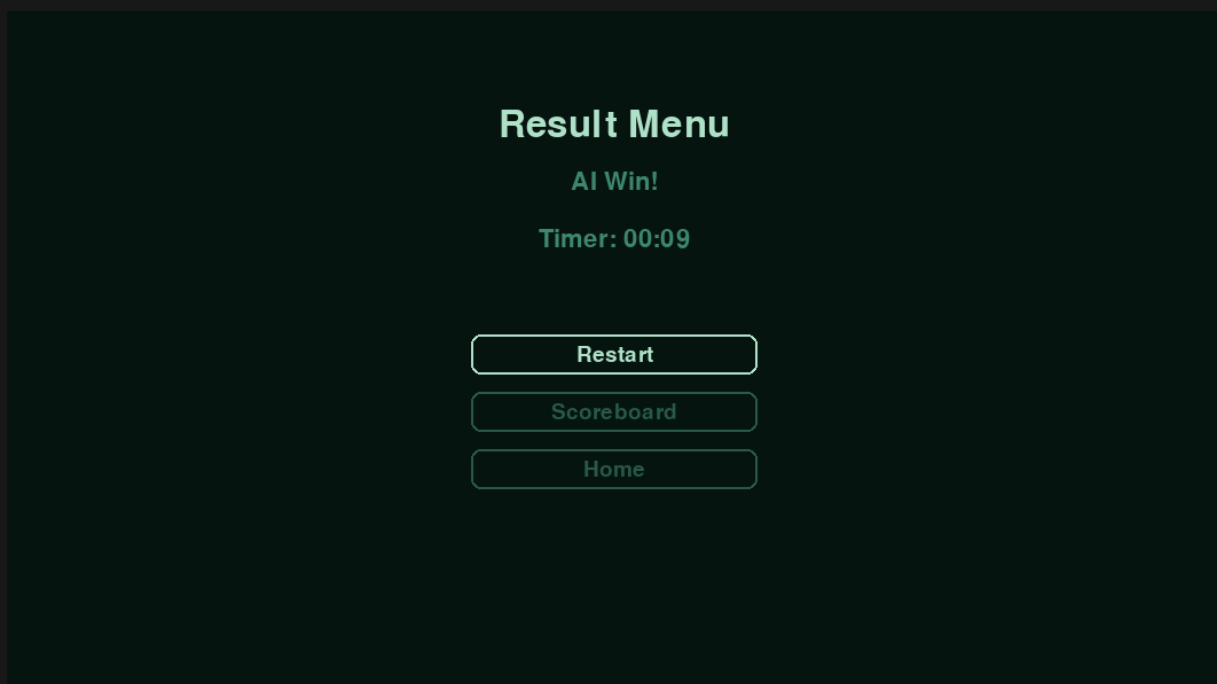
Level: 1

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Pause Menu



Result Menu



Scoreboard Menu



Score Board Menu

99:99:99

99:99:99

99:99:99

99:99:99

99:99:99

Wins: 0

Loses: 0

Test Cases

Fields	Values
Test Case ID	TC01
Test Scenario	Testing Select Difficulty button which takes user to the difficulty menu, allowing user to select difficulty
Preconditions	Current menu: Main menu
Actions	1. Press Enter key or left mouse click on Select Difficulty button.2. Press Enter key or left mouse click on either Easy, Medium or hard button
Expected Results	Takes user to main menu and generates a new maze
Actual Results	As Expected
Tested By	Muhammad Zaid(BC220407371)
Pass / Fail	Pass

Fields	Values
Test Case ID	TC02
Test Scenario	Testing Play button which starts the game
Preconditions	Current menu: Main menu
Actions	Press Enter key or left mouse click on Play button
Expected Results	Game starts
Actual Results	As Expected
Tested By	Muhammad Zaid(BC220407371)
Pass / Fail	Pass

Fields	Values
Test Case ID	TC03
Test Scenario	Testing navigation and making sure player moves in the directions corresponding to the arrow keys on the keyboard
Preconditions	Current Game State: Playing and there is no wall in front of player
Actions	Press arrow keys
Expected Results	Player should move in the corresponding direction
Actual Results	As Expected
Tested By	Muhammad Zaid(BC220407371)
Pass / Fail	Pass

Fields	Values
Test Case ID	TC04
Test Scenario	Testing Pause button which pauses the game while it is being played
Preconditions	Current Game State: Playing
Actions	Press Esc key or P key or left mouse click on Pause button
Expected Results	Takes user to Pause menu
Actual Results	As Expected
Tested By	Muhammad Zaid(BC220407371)
Pass / Fail	Pass

Fields	Values
Test Case ID	TC05
Test Scenario	Testing Resume button which resumes the game without changing anything
Preconditions	Current menu: Pause menu
Actions	Press Enter key or left mouse click on Resume button
Expected Results	Takes user to back to playing state
Actual Results	As Expected
Tested By	Muhammad Zaid(B220407371)
Pass / Fail	Pass

Fields	Values
Test Case ID	TC06
Test Scenario	Testing Restart button which resets the level being currently played
Preconditions	Current menu: Pause menu or Result menu
Actions	Press Enter key or left mouse click on Restart button
Expected Results	Reset everything on playing screen except for maze structure
Actual Results	As Expected
Tested By	Muhammad Zaid(BC220407371)
Pass / Fail	Pass

Fields	Values
Test Case ID	TC07
Test Scenario	Testing Scoreboard button which shows the wins, losses and completion times
Preconditions	Current Menu: Main menu or Result menu
Actions	Press Enter key or left mouse click on Scoreboard button
Expected Results	Takes user to Scoreboard menu
Actual Results	As Expected
Tested By	Muhammad Zaid(BC220407371)
Pass / Fail	Pass

Fields	Values
Test Case ID	TC08
Test Scenario	Testing Next button which advances to the next level
Preconditions	Current Menu: Result Menu
Actions	Press Enter key or left mouse click on Next button
Expected Results	Advances user to next level
Actual Results	As Expected
Tested By	Muhammad Zaid(BC220407371)
Pass / Fail	Pass

Fields	Values
Test Case ID	TC09
Test Scenario	Testing Home button which takes to the main menu
Preconditions	Current menu: Pause menu or Result menu or Difficulty menu
Actions	Press Enter key or left mouse click on Home button
Expected Results	Takes user to main menu
Actual Results	As Expected
Tested By	Muhammad Zaid(BC220407371)
Pass / Fail	Pass

Fields	Values
Test Case ID	TC10
Test Scenario	Testing Exit button which terminates the game
Preconditions	Current menu: Main menu
Actions	Press Enter key or left mouse click on Exit button
Expected Results	Saves data and terminate the game
Actual Results	As Expected
Tested By	Muhammad Zaid(BC220407371)
Pass / Fail	Pass